

AI Companion Testing Plan

Test scenarios for the AI Companion Prototype on the `AI-Companion-Prototype` branch.

Setup

- Open `Lvl_FirstPerson` (or your test level)
 - Confirm `BP_Companion` is placed near `PlayerStart`
 - Confirm 3-4 `BP_EnergyBase` placed around the level (some near walls/cover)
 - Confirm `NavMeshBoundsVolume` covers the playable area (press `P` to visualise — green = walkable)
 - Open the **Output Log** (Window > Output Log) — filter by `LogCompanion`, `LogEnemy`, `LogExtraction`
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Test 1 — Follow Behaviour

Goal: Companion stays in formation at a comfortable distance.

Step	Expected
Walk forward	Companion follows behind/right at ~350cm
Turn 360°	Companion does NOT spin around you (uses velocity, not look direction)
Stand still	Companion stops, holds position
Sprint away	Companion sprints to catch up, then walks when close
Walk into companion	Companion holds ground, doesn't push through you

Failure signs: jittering, getting inside player, sprinting at full speed all the time.

Test 2 — Enemy Detection & Combat

Goal: Companion sees enemy, engages, fires accurately over time.

Step	Expected
Walk into enemy line of sight	Companion immediately rotates toward enemy
Watch debug lines	Yellow = miss, Red = hit. Should fire in 1.5s bursts with 0.5s pauses
Stay aiming at same enemy	Inaccuracy decreases (look at red lines clustering tighter)
Magazine empties	Companion stops firing, reloads (~2s), resumes

Step	Expected
Enemy dies	Combat ends, returns to follow
Multiple enemies	Picks closest, switches if closer one appears

Output Log: Companion hit BP_EnemyBase for 25.0 damage

Test 3 — Cover System

Goal: Companion uses environment for cover before engaging.

Step	Expected
Place enemy near a wall/pillar	Companion runs to cover spot first
Engage from open area (no cover)	Companion engages from current position (graceful fallback)
Watch in-game	Companion ends up between enemy and a wall

Failure signs: running into enemy, ignoring cover, freezing.

Test 4 — Revive (THE MONEY SHOT)

Goal: Companion drops everything to revive you when downed.

Step	Expected
Press H ~4 times to reach DBNO	Player enters DBNO state
Watch companion	Stops current action immediately (mid-combat or follow)
Companion sprints to you	Max speed, ignores enemies
Companion arrives within 200cm	Stops, faces you, holds 4s
After 4s	Player exits DBNO, restored to 30% HP
Post-revive	Companion returns to combat/follow

Output Log:

- Companion starting revive sequence for BP_ExtractionCharacter
 - Companion revived BP_ExtractionCharacter
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Test 5 — Damage & Death

Goal: Companion takes damage and dies properly.

Step	Expected
Let enemies shoot companion	HP/shield decreases (Output Log: Companion took X damage – HP: Y / Shield: Z)
Shield depletes first, then HP	Standard health flow
Companion HP hits 0	Stops moving/firing, collision off, destroyed after 3s

Test 6 — Behaviour Priority

Goal: Higher-priority behaviours interrupt lower ones.

Scenario	Expected
Mid-combat, you go DBNO	Combat aborts instantly, companion sprints to revive
While following, enemy appears	Follow aborts, companion engages
Mid-revive, enemy fires at companion	Revive continues (highest priority)

Known Limitations (prototype scope)

- Cylinder mesh only — no animations
- No squad coordination (single companion)
- No voice lines / barks
- Companion ignores its own DBNO (dies immediately when HP=0)
- No equipment switching

Bug Report Template

When something breaks, capture:

1. What you were doing
2. What companion was doing before
3. Output Log filtered by LogCompanion/LogEnemy
4. Screenshot if visual